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Sample Question Set - Unit 6

Course Code: CSE276
Course Title: ARTIFICIAL INTELLIGENCE FOUNDATIONS
Course Outcome: CO6
Paper: Sample Multiple Choice Questions - ANSWER KEY
Maximum Marks: 40 x 1 = 40

Unit: 6
Unit Title: AI Development Ecosystem and Future Trends
Unit Mapped RBT: L3 (Apply)
Total Questions: 40
Suggested Duration: 60 Minutes

Answer Grid (Key)

Q. No.	Answer	Q. No.	Answer	Q. No.	Answer	Q. No.	Answer
Q1	A	Q11	D	Q21	A	Q31	C
Q2	B	Q12	B	Q22	C	Q32	A
Q3	C	Q13	D	Q23	B	Q33	A
Q4	D	Q14	C	Q24	D	Q34	D
Q5	B	Q15	B	Q25	C	Q35	B
Q6	D	Q16	A	Q26	D	Q36	C
Q7	A	Q17	B	Q27	A	Q37	C
Q8	C	Q18	A	Q28	B	Q38	B
Q9	C	Q19	D	Q29	D	Q39	D
Q10	A	Q20	C	Q30	B	Q40	A

Multiple Choice Questions (40 x 1 = 40 Marks)

Q1. Name the Python library used for numerical arrays and vectorised arithmetic.

A) NumPy <-- correct
B) Matplotlib
C) LangChain
D) Ollama

Answer: A. NumPy provides the array type on which almost the whole Python data and machine learning ecosystem is built.

Q2. Identify the deep learning framework originally developed at Google.

A) PyTorch
B) TensorFlow <-- correct
C) scikit-learn
D) Hugging Face

Answer: B. TensorFlow and PyTorch are the two frameworks named on the syllabus; PyTorch came from Meta and is the more common in research.

Q3. State what Ollama is used for.

A) Publishing datasets for other people to use
B) Storing embeddings for similarity search
C) Running open models locally on your own machine <-- correct
D) Drawing charts from a table of results

Answer: C. Ollama packages an open model so it can be pulled and served on a local machine, which keeps the data on the premises.

Q4. Name the tool used to connect a language model together with retrieval and other components into one pipeline.

A) NumPy
B) TensorFlow
C) Matplotlib
D) LangChain <-- correct

Answer: D. LangChain supplies the plumbing between a model, a retriever, a prompt template and any tools the application calls.

Q5. Identify the similarity measure most commonly used inside a vector database.

A) Manhattan distance
B) Cosine similarity <-- correct
C) Hamming distance
D) Edit distance

Answer: B. Embeddings encode meaning as direction, so the angle between two vectors is what matters and their magnitude is not informative.

Q6. State the Python library that provides the classical machine learning algorithms.

A) TensorFlow
B) Ollama

C) LangChain

D) scikit-learn <-- correct

Answer: D. scikit-learn covers regression, classification, clustering and the preprocessing around them with one consistent interface.

Q7. Explain what an approximate nearest neighbour index gives up in exchange for its speed.

A) It may occasionally miss the very closest match <-- correct

B) It may return results in the wrong language

C) It may store the vectors in the wrong order

D) It may require the model to be retrained

Answer: A. Exact search must compare the query with every vector. The approximate index searches a narrowed region, which is far quicker and almost always as good.

Q8. Distinguish TensorFlow and PyTorch from scikit-learn by what each is used for.

A) The first two build classical models, the third networks

B) All three of them build only neural networks

C) The first two build neural networks, the third classical models <-- correct

D) All three of them build only classical models

Answer: C. Choosing the right library for the task is part of the workflow: a tabular problem rarely needs a deep learning framework at all.

Q9. Describe what a virtual environment contributes to a Python project.

A) It speeds up the training of the model

B) It encrypts the data the project uses

C) It isolates the project's library versions from others <-- correct

D) It converts the code into another language

Answer: C. Two projects can then depend on different versions of the same library without conflict, which is the first step towards a reproducible result.

Q10. Interpret the following requirement and name the component it calls for: an assistant must run with no data leaving the organisation's premises.

A) An open model served locally <-- correct

B) A hosted commercial model over an API

C) A public dataset from a model hub

D) A shared vector store in the cloud

Answer: A. Local serving is the only arrangement in which the prompt and the retrieved documents never cross the organisation's boundary.

Q11. Predict what happens when the passages used for retrieval are made far too large.

A) The vector store refuses to accept them

B) The embedding model produces shorter vectors

C) The retrieval becomes exact rather than approximate

D) Each retrieved passage carries much irrelevant text <-- correct

Answer: D. A whole chapter embedded as one vector has a blurred meaning, so matching is poorer and the context window fills with material that does not bear on the question.

Q12. Recognize the stage described: a held-out set is consulted once in order to estimate how the system will behave in service.

A) Data collection

B) Evaluation before release <-- correct

C) Problem definition

D) Monitoring after release

Answer: B. Consulting the held-out set more than once turns it into part of the development data, and the estimate it gives stops being honest.

Q13. Explain why the embedding dimension must match between the stored vectors and the query.

A) The database will silently pad the shorter vector

B) The model will retrain itself to fit the store

C) The dimension has no bearing on the search

D) Similarity cannot be computed between vectors of different lengths <-- correct

Answer: D. This is why replacing the embedding model requires the whole store to be rebuilt, not merely the new documents to be added.

Q14. Summarize what sustainable Artificial Intelligence is concerned with.

A) The length of time a model remains accurate

B) The number of users a system can support

C) The energy and hardware that training and serving consume <-- correct

D) The licence under which a model is released

Answer: C. It covers the choice of model size, the reuse of pre-trained weights rather than training afresh, and the efficiency of the serving arrangement.

Q15. Distinguish a model card from the model weights themselves.

A) The weights document the model, the card is the model

B) The card documents the model, the weights are the model <-- correct

C) Both of the two are documentation of the model

D) Both of the two are numeric parameter files

Answer: B. A model card records the intended use, the training data, the known limitations and the evaluation results, which is what lets a user judge suitability.

Q16. Explain why the Artificial Intelligence development workflow is drawn as a loop rather than a straight line.

A) Monitoring sends the project back to earlier stages <-- correct

B) Every stage has to be repeated a fixed number of times

C) The stages can be carried out in any order at all

D) The workflow has no defined end point of any kind

Answer: A. Incoming data drifts and requirements change, so a deployed system returns to data collection and retraining rather than being finished at release.

Q17. Describe what LangChain adds to a plain call to a language model.

A) A faster implementation of the model itself

B) Orchestration of prompts, retrieval and tool calls <-- correct

C) A larger context window for the model

D) A guarantee that answers are factually correct

Answer: B. It supplies the structure around the model. The model's own behaviour, including its tendency to hallucinate, is entirely unchanged by it.

Q18. Predict the difficulty of reproducing a result when the versions of the libraries used were never recorded.

A) A later version may change the behaviour silently <-- correct

B) The code will refuse to run at all

C) The data will be deleted automatically

D) The model weights will change by themselves

Answer: A. Defaults and algorithms change between releases, so the same script can give a different answer months later with nothing in the code altered.

Q19. Explain why reusing a pre-trained model is more sustainable than training an equivalent one from the beginning.

A) Pre-trained models are stored in fewer bytes

B) Pre-trained models need no hardware to run

C) Pre-trained models require no evaluation

D) The largest share of the energy has already been spent once <-- correct

Answer: D. Pre-training dominates the energy cost of a model's whole life. Adapting an existing one spreads that single cost across many downstream uses.

Q20. Interpret the claim that a system has reached Artificial General Intelligence because it writes fluent essays.

A) The claim is correct, since writing requires reasoning

B) The claim is correct, since essays cover many subjects

C) Fluency in one task is not breadth across all tasks <-- correct

D) The claim cannot be judged without the source code

Answer: C. General intelligence is defined by transfer across arbitrary domains. Excellent performance within one bounded task is exactly what narrow AI already is.

Q21. Solve the following: 20,000 passages are embedded as vectors of 384 numbers, each stored in 4 bytes. Determine the memory the vectors occupy.

A) About 30.7 MB <-- correct

B) About 7.7 MB

C) About 76.8 MB

D) About 3.1 MB

Answer: A. 20,000 x 384 x 4 = 30,720,000 bytes, which is approximately 30.7 MB.

Q22. Compute the monthly cost of 15,000 queries charged at 0.004 dollars each.

A) 6 dollars

B) 600 dollars

C) 60 dollars <-- correct

D) 150 dollars

Answer: C. 0.004 x 15,000 = 60 dollars, the figure to weigh against the cost of running an open model on your own hardware.

Q23. Apply the development workflow and determine the stage at which the requirement and the success criteria are agreed.

A) Data preparation

B) Problem definition <-- correct

C) Evaluation

D) Monitoring

Answer: B. Settling what the system is for, and how success will be judged, before anything is built is what prevents disputes surfacing only at the end.

Q24. Use the idea of a vector database to determine what is stored when 800 documents are added to a retrieval assistant.

A) One embedding for each whole document only

B) The complete text of every document alone

C) The parameters of the language model itself

D) One embedding per passage, with its source recorded <-- correct

Answer: D. Keeping the source alongside each vector is what allows the finished answer to cite the passage it was drawn from.

Q25. Examine a retrieval assistant that returns passages on entirely the wrong subject, and determine what to check first.

A) The temperature setting of the language model

B) The number of parameters the model holds

C) The passages, the embedding model and the index <-- correct

D) The programming language of the application

Answer: C. Retrieval faults lie in the retrieval half of the pipeline. The generator can only work with the passages it is handed.

Q26. Compare running a seven-billion-parameter model on a laptop with calling a hosted service, on latency and on cost.

A) Local costs per query and is usually faster

B) The two are identical on both latency and cost

C) Hosted costs nothing per query and is slower

D) Local costs nothing per query but is usually slower <-- correct

Answer: D. The hosted service runs on server-grade accelerators, so it usually answers faster, but it charges for every query and the data leaves the premises.

Q27. Solve the following: a model requires 14 GB of memory and the available accelerator has 8 GB. Determine the appropriate action.

A) Quantise the model or choose a smaller one <-- correct

B) Increase the temperature setting

C) Reduce the size of the context window

D) Add more documents to the vector store

Answer: A. Quantisation to fewer bytes per parameter, or selecting a smaller variant, are the two routes to fitting the weights into the memory available.

Q28. Apply Hugging Face to a requirement for a pre-trained model and determine what it supplies.

A) The hardware on which the model will run

B) The weights, the model card and the code to load them <-- correct

C) A guarantee that the model is free of bias

D) The private data of the organisation using it

Answer: B. The hub distributes the artefacts and their documentation. Judging whether a model suits the task remains the user's responsibility.

Q29. Use the difference between exact and approximate search to determine which suits a store of ten million vectors.

A) Exact search, since it is always faster

B) Neither, since ten million vectors cannot be stored

C) Either, since the two cost the same at this size

D) Approximate search, since exact search is too slow <-- correct

Answer: D. Exact search compares the query with all ten million vectors on every request. An approximate index narrows that to a small candidate set.

Q30. Compute the number of vectors held when 500 documents are each split into 20 passages.

A) 500

B) 10,000 <-- correct

C) 520

D) 20

Answer: B. Each passage becomes one vector: 500 x 20 = 10,000 entries in the store.

Q31. Examine a deployment whose answers grow slower as the document store expands, and determine the remedy.

A) Reduce the number of documents in the store

B) Raise the temperature of the language model

C) Build a vector index in place of the linear scan <-- correct

D) Retrain the language model on the documents

Answer: C. Comparing the query with every stored vector costs time in proportion to the store. An index keeps the search roughly constant as the store grows.

Q32. Apply version pinning to a project and determine what it prevents.

A) A silent change in behaviour after a library update <-- correct

B) A rise in the cost of each hosted query

C) A leak of personal data from the vector store

D) A fall in the accuracy of the embedding model

Answer: A. Pinning the versions of the libraries and the model weights is what makes a result obtained today reproducible in six months.

Q33. Use the development workflow to determine what should be done immediately before a system is released.

A) Evaluate on the held-out set and record the result <-- correct

B) Collect a further batch of training data

C) Increase the number of model parameters

D) Delete the documentation of the project

Answer: A. The recorded evaluation is the baseline against which the monitoring after release is compared, so drift can be recognised when it appears.

Q34. Compare a model card with an evaluation report on what each tells a prospective user.

A) The report states intended use, the card the performance

B) Both of the two state only the measured performance

C) Both of the two state only the intended use

D) The card states intended use and limits, the report the measured performance <-- correct

Answer: D. A user needs both: whether the model was meant for this task at all, and how well it actually did on data resembling their own.

Q35. Solve the following: an embedding model produces 768 numbers for each passage, stored at 4 bytes each. Determine the memory for 2,000 passages.

A) About 1.5 MB

B) About 6.1 MB <-- correct

C) About 24.6 MB

D) About 0.8 MB

Answer: B. 2,000 x 768 x 4 = 6,144,000 bytes, which is approximately 6.1 MB.

Q36. Apply the idea of sustainable Artificial Intelligence and determine which change most reduces the energy used in serving a model.

A) Adding more documents to the store

B) Raising the temperature of the model

C) Serving a smaller or quantised model <-- correct

D) Writing a longer system prompt

Answer: C. Inference energy is dominated by the size of the model and the number of queries, so shrinking the model is the most direct lever.

Q37. Examine a project that can no longer reproduce a result obtained last month, and determine the most likely cause.

A) The evaluation set was too large to store

B) The vector store used cosine similarity

C) The library versions or the random seed were not recorded <-- correct

D) The model card was written after release

Answer: C. Unpinned dependencies and an unrecorded seed are the two commonest reasons a script gives a different answer on a later run.

Q38. Use the definition of Artificial General Intelligence to judge a system that translates fluently between sixty languages.

A) It qualifies, since sixty languages is very broad

B) It remains narrow, however impressive its range <-- correct

C) It qualifies, since translation requires understanding

D) It cannot be judged without seeing its parameters

Answer: B. Breadth within one task is not breadth across tasks. The system cannot transfer its competence to planning, vision or reasoning about the physical world.

Q39. Demonstrate how an assistant grounded in internal documents is assembled from the tools of the ecosystem.

A) A single language model fine-tuned on every document

B) A relational database queried with exact keywords

C) A rule-based expert system with a knowledge base

D) An embedding model and vector store for retrieval, a served model to answer, and an orchestrator between them <-- correct

Answer: D. Retrieval keeps the documents outside the model so they can be updated and cited, and the orchestration layer joins the retriever to the generator.

Q40. Relate the number of stored passages to the need for an index rather than a straightforward comparison against every vector.

A) The larger the store, the greater the need for an index <-- correct

B) The larger the store, the less the need for an index

C) The size of the store has no bearing on the need

D) An index is required even for a single passage

Answer: A. A few thousand vectors can be scanned directly without a noticeable delay. Beyond that the cost of the scan starts to dominate the response time.

----- End of Sample Question Set -----